

Swathi R

Senior Data Engineer

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PROFESSIONAL SUMMARY

- Over **8+ years of IT experience** in Big Data, Data Engineering, ETL, and Cloud technologies with expertise in designing and developing scalable data processing solutions.
- Strong hands-on experience with **Hadoop Ecosystem** including HDFS, MapReduce, Hive, Pig, YARN, Sqoop, Flume, HBase, Impala, Oozie, ZooKeeper, Kafka, and PySpark.
- Extensive experience in **Apache Spark** (Spark Core, Spark SQL, Spark Streaming, MLlib, GraphX, DataFrames, RDDs) for large-scale batch and real-time data processing.
- Expertise in building scalable **ETL/ELT pipelines** using **PySpark, AWS Glue, Azure Databricks, and Kafka** for processing high-volume structured and unstructured data.
- Strong experience with **AWS** (S3, EMR, Glue, Redshift, Lambda, IAM, EC2) and **Azure Databricks** for cloud-native data engineering solutions.
- Proficient in SQL, data modeling (Star & Snowflake Schema), data warehousing, and performance tuning across Oracle, Hive, Cassandra, and HBase.
- Experience supporting **AI/ML workflows** through feature engineering, training data validation, data quality checks, schema validation, and anomaly detection using PySpark and SQL.
- Experienced in Agile/Scrum environments with excellent analytical, problem-solving, and cross-functional collaboration skills.

TECHNICAL SKILLS

Programming Languages

Python, SQL, Shell Scripting,

Big Data Technologies

Hadoop, HDFS, MapReduce, Hive, Apache Spark, PySpark, Spark SQL, Kafka, Sqoop, Flume, HBase, YARN, Oozie, ZooKeeper

Cloud Platforms

AWS (S3, EMR, Glue, Lambda, Redshift, IAM, EC2, CloudWatch)

Microsoft Azure (Azure Databricks, Azure Data Factory, Azure Storage, Azure Key Vault)

Data Engineering & ETL

AWS Glue, Azure Data Factory (ADF), Apache Airflow, Delta Lake, Delta Live Tables (DLT), ETL/ELT Pipelines, Data Ingestion, Data Migration, REST APIs

Databases

Oracle, SQL Server, PostgreSQL, MySQL, Cassandra, HBase

Data Warehousing

Snowflake, Amazon Redshift, Hive, Star Schema, Snowflake Schema

Data Processing & Analytics

Data Cleansing, Data Transformation, Data Validation, Data Quality, Data Profiling, Data Modeling, Feature Engineering, Batch Processing, Real-Time Data Processing

Machine Learning Support

AI/ML Data Preparation, Training Data Validation, Schema Validation, Data Quality Checks, Anomaly Detection

Reporting & Visualization

Power BI, Tableau, Microsoft Excel

Version Control & DevOps

Git, GitHub, GitLab, Jenkins, Azure DevOps, Terraform

Operating Systems

Linux, Unix

Methodologies

Agile (Scrum), SDLC

Senior Data Engineer

Odyssey Group

🏠 Mar 2025 – Present 📍 Stamford, CT

Project Description: Designed and developed cloud-native data engineering solutions to support enterprise reporting, advanced analytics, and AI/ML initiatives. Built scalable batch and real-time data pipelines using **PySpark, Apache Spark, Kafka, AWS Glue, Azure Databricks, and SQL** to process high-volume structured and semi-structured data while ensuring data quality, performance, and reliability.

- Designed and implemented scalable Delta Lake data pipelines using Azure Databricks and PySpark to process enterprise-scale structured and semi-structured data.
- Developed Apache Airflow DAGs to orchestrate complex ETL workflows, improving pipeline automation and reducing manual intervention.
- Built incremental data loading frameworks using Delta Live Tables (DLT) and optimized data processing for high-volume datasets.
- Developed and maintained Medallion Architecture (Bronze, Silver, Gold) to improve data quality and support enterprise analytics.
- Implemented Slowly Changing Dimensions (SCD Type 1 & Type 2) and Change Data Capture (CDC) techniques for maintaining historical data.
- Loaded curated datasets into Snowflake to support business intelligence, regulatory reporting, and self-service analytics.
- Configured Unity Catalog to implement centralized data governance, access control, and metadata management across Databricks workspaces.
- Optimized PySpark applications using partitioning, caching, broadcast joins, and adaptive query execution to improve Spark job performance.
- Implemented enterprise data quality frameworks, schema validation, and automated reconciliation processes using PySpark and SQL.
- Developed real-time streaming pipelines using Apache Kafka and integrated streaming data into cloud-based analytics platforms.
- Implemented REST API-based integrations to ingest data from enterprise applications into the cloud data platform.
- Managed cloud-based data solutions using AWS S3, Glue, IAM, CloudWatch, and Azure Databricks, ensuring secure and scalable data processing.
- Automated CI/CD deployments using Azure DevOps, Git, and Jenkins, supporting version control and seamless production releases.
- Collaborated with Data Scientists, Business Analysts, and cross-functional Agile teams to deliver reliable datasets for AI/ML models and enterprise reporting.

Data Engineer

Swift

🏠 Jan 2018 – Dec 2024 📍 Manassas, VA

Project Description: Designed and implemented enterprise data warehouse solutions using Hadoop and Spark technologies to process large volumes of transactional and operational data. Developed batch ETL pipelines using Hive, Sqoop, Spark, and AWS services to support regulatory reporting, business intelligence, and enterprise analytics.

- Designed and developed scalable ETL/ELT pipelines using PySpark, Spark SQL, Hive and Sqoop and Spark on YARN clusters for large-scale data processing.
- Built real-time data ingestion pipelines using Apache Kafka to process streaming data from multiple enterprise systems.
- Developed and optimized PySpark applications for data cleansing, transformation, aggregation, and validation.
- Developed Hive external and managed tables to support enterprise reporting and analytical workloads.
- Designed optimized SQL queries and implemented data models to improve reporting performance and analytics.
- Imported enterprise data using Sqoop from Oracle and SQL Server into Hadoop for downstream analytics.
- Developed Spark and Hive ETL workflows to cleanse, transform, and aggregate enterprise data.
- Optimized Spark jobs using partitioning, caching, and broadcast joins to improve application performance.
- Worked extensively with AWS S3, EMR, Glue, Redshift, IAM, and CloudWatch for cloud-based data engineering solutions.

- Collaborated with Data Scientists, Business Analysts, and cross-functional teams in an Agile/Scrum environment to deliver business solutions.
- Automated deployment and version control using Git, Jenkins, ensuring reliable CI/CD processes.
- Monitored production data pipelines, resolved performance issues, and performed root cause analysis to maintain high system availability.

Data Analyst

Magellan Health

📅 Mar 2017 – Jan 2018 📍 Frisco, TX

Project Description: Developed analytical solutions to support healthcare operations, member services, claims analysis, and business reporting. Collected, transformed, and analyzed healthcare data from multiple sources using SQL, Python, and ETL processes to generate actionable insights, improve operational efficiency, and support data-driven decision-making while ensuring HIPAA compliance.

- Collected, cleaned, and analyzed healthcare data using SQL and Python to support business reporting and operational analysis.
- Developed interactive dashboards and reports using Power BI and Tableau to monitor claims, member enrollment, and healthcare performance metrics.
- Wrote complex SQL queries, stored procedures, and views to extract, transform, and validate data from multiple source systems.
- Created ad-hoc reports and KPI dashboards to support healthcare claims, provider performance, and member utilization analysis.
- Performed data profiling, validation, and quality checks to ensure accuracy, consistency, and completeness of healthcare data.
- Collaborated with business users and stakeholders to gather reporting requirements and deliver analytical solutions.
- Supported ETL processes by validating source-to-target mappings and troubleshooting data quality issues.
- Analyzed healthcare claims, member eligibility, and provider data to identify trends and support business decisions.
- Automated recurring reports and data preparation tasks using Python, reducing manual effort and improving reporting efficiency.
- Ensured compliance with HIPAA guidelines while handling sensitive healthcare and patient-related information.
- Worked in an Agile/Scrum environment, supporting UAT, documentation, and production issue resolution.

EDUCATION

Master's in Computer Science

Kent State University 📍 Ohio 📅 2016

Bachelors in Information Technology

JNTUH 📍 India 📅 2014